

Apple juice improved behavioral but not cognitive symptoms in moderate-to-late stage Alzheimer's disease in an open-label pilot study.

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Preclinical studies demonstrate that apple juice exerts multiple beneficial effects including reduction of central nervous system oxidative damage, suppression of Alzheimer's disease (AD) hallmarks, improved cognitive performance, and organized synaptic signaling. Herein, we initiated an open-label clinical trial in which 21 institutionalized individuals with moderate-to-severe AD consumed 2 4-oz glasses of apple juice daily for 1 month. Participants demonstrated no change in the Dementia Rating Scale, and institutional caregivers reported no change in Alzheimer's Disease Cooperative Study (ADCS)-Activities of Daily Living (ADL) in this brief study. However, caregivers reported an approximate 27% ($P < .01$) improvement in behavioral and psychotic symptoms associated with dementia as quantified by the Neuropsychiatric Inventory, with the largest changes in anxiety, agitation, and delusion. This pilot study suggests that apple juice may be a useful supplement, perhaps to augment pharmacological approaches, for attenuating the decline in mood that accompanies progression of AD, which may also reduce caregiver burden.